MSc Thesis Title

By

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Declaration

I declare that the work presented in this thesis has not been submitted for any Degree or Diploma at this, or any other University and that the work described.

I confirm that all sources of information used in the preparation of this dissertation have been appropriately acknowledged and referenced in accordance with the required academic standards. Any assistance received during the completion of this work has been fully acknowledged.

I further declare that informed consent was obtained from the participant and/or her legal guardian prior to the commencement of the case study. All data were collected and handled in accordance with ethical guidelines, maintaining participant confidentiality and anonymity throughout the study.

Signed:……………………………………………………… Date:………………..

Acknowledgements

I would like to express my sincere gratitude to my academic supervisor for their continuous guidance, support, and valuable feedback throughout the completion of this dissertation. Their expertise and encouragement greatly contributed to the successful completion of this study.

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Special thanks are extended to the participant and her family for their cooperation, trust, and willingness to participate in this case study. Their contribution made this research possible.

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I would also like to acknowledge the invaluable learning experiences gained throughout my placement, which played a significant role in the development of both my professional and academic skills.

Working closely with individuals with complex health and developmental needs provided me with a deeper understanding of the challenges faced by people with intellectual disabilities and neurological conditions such as epilepsy.

This experience enhanced my appreciation of the importance of individualized exercise prescription, patient-centred care, and multidisciplinary collaboration in achieving positive health outcomes.

Observing and participating in the assessment, planning, implementation, and monitoring of exercise interventions allowed me to apply theoretical knowledge acquired during the MSc programme in a practical setting.

Abstract

Individuals with intellectual disability often experience reduced physical activity levels, lower functional capacity, and an increased prevalence of secondary health conditions. The presence of epilepsy may further complicate participation in exercise programmes due to concerns regarding safety, supervision, and seizure management. The purpose of this case study was to evaluate the implementation of an individualized exercise programme for a female participant with intellectual disability and mild epilepsy within a community-based setting.

A comprehensive assessment was conducted to evaluate the participant's functional capacity, mobility, balance, muscular strength, and ability to perform activities of daily living. Based on the findings, an individualized exercise intervention was developed, focusing on improving physical fitness, functional independence, and overall quality of life while ensuring appropriate safety measures related to epilepsy management.

The intervention consisted of supervised aerobic, resistance, balance, and flexibility exercises performed over the placement period. Progress was monitored through observational assessments and repeat functional testing. Improvements were observed in functional mobility, exercise tolerance, balance, and confidence during physical activity participation. The participant demonstrated good adherence to the exercise programme and tolerated the intervention without adverse events.

The findings of this case study suggest that appropriately prescribed and supervised exercise can be a safe and effective intervention for individuals with intellectual disability and mild epilepsy. Exercise programmes tailored to individual abilities and health needs may contribute to improved physical function, independence, and overall well-being. Further research involving larger populations is required to establish evidence-based exercise guidelines for this population.

In addition to the physical outcomes, the intervention aimed to encourage regular participation in structured exercise and promote positive attitudes towards physical activity. Individuals with intellectual disability frequently encounter barriers to exercise participation, including limited access to suitable programmes, communication difficulties, and reduced confidence in performing physical tasks.

Therefore, the exercise programme was designed to be enjoyable, achievable, and adapted to the participant's cognitive and physical abilities. Particular consideration was given to seizure risk management, exercise intensity monitoring, and environmental safety throughout all sessions.

Regular supervision and individualized progression ensured that the participant could engage safely while maximizing the potential benefits of exercise training.

Furthermore, the intervention emphasized the development of functional skills that could enhance independence in everyday activities and support long-term health maintenance

List of commonly used abbreviations

Abbreviation	Meaning
ACSM	American College of Sports Medicine
BMI	Body Mass Index
ID	Intellectual Disability
ICF	International Classification of Functioning
ROM	Range of Motion
6MWT	Six-Minute Walk Test
WHO	World Health Organization
DSM-5-TR	Diagnostic and Statistical Manual of Mental Disorders
FITT	Frequency, Intensity, Time and Type
ADL	Activities of Daily Living

TABLE OF CONTENTS

Appendices